

中国科学院大学
University of Chinese Academy of Sciences

Generative Models & AI4S Applications

Lecture 5 Latent Variable Models I

Fall 2026

Liu Yunfan (刘云帆)

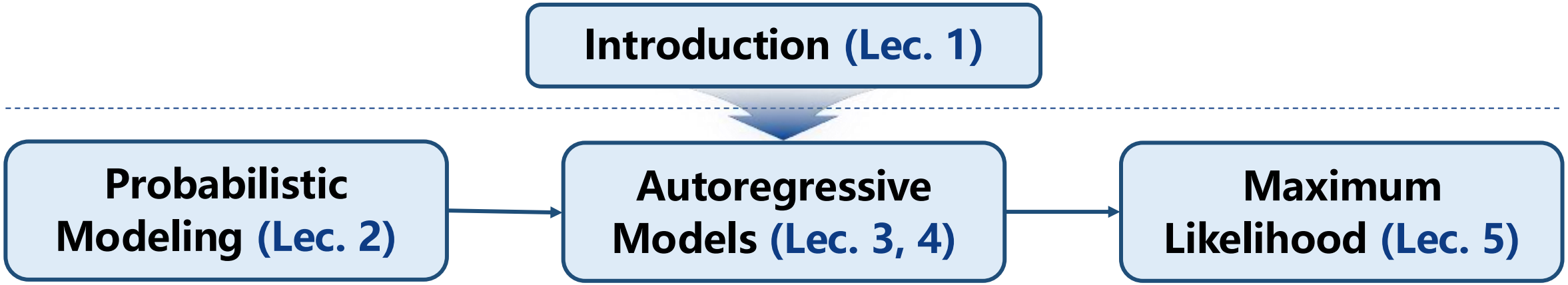
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Announcements

- Assignment 1 **due on October 18th**

- Slides of this lecture subject to future changes and will be updated online

Roadmap



An overview of the pipeline of learning a generative model

Recap

- **Lecture 2: Probabilistic Modeling → Representing P_θ**
  - **Chain Rule:** Joint Distribution → Conditional Distributions (too expensive)
  - **Bayesian Network:** Simplify by assuming conditional independence
  - **Neural Network:** Simplify by parameterization
- **Lecture 3 & 4: Autoregressive Models → A Class of P_θ**
  - AR-ness / ordering, teacher forcing / exposure bias
  - Different models and architectures

$$p(\mathbf{x}) = \prod_{i=1}^{n} p(x_i \mid \mathbf{x}_{<i})$$

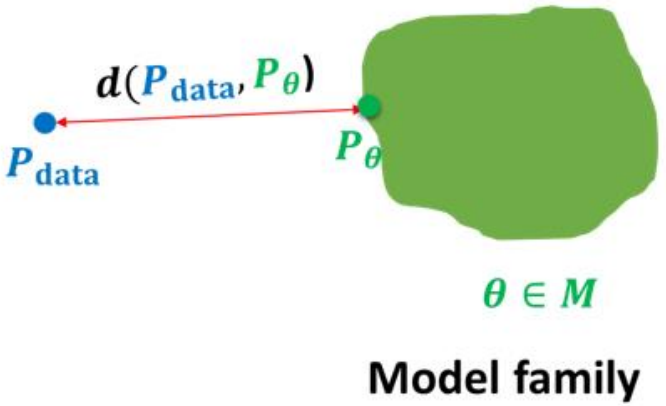
- **Lecture 5: Maximum Likelihood Learning → Learning θ**
  - KL Divergence, Monte Carlo Estimation, MLE

Recap

Images of dogs



$x_i \sim P_{\text{data}}$
$i = 1, 2, \ldots, n$



Dog Image Generator

- Machine learning methods:
  - **Model Family**
  - **Learning Objective**
  - **Optimization**

$$\theta^* = \arg\min_\theta \mathbb{E}_{x \sim p_{\text{data}}(x)}[\mathcal{L}(x; \theta)]$$

- Functions of generative models:
  - **Generation**
  - **Density Estimation**
  - **Representation Learning**

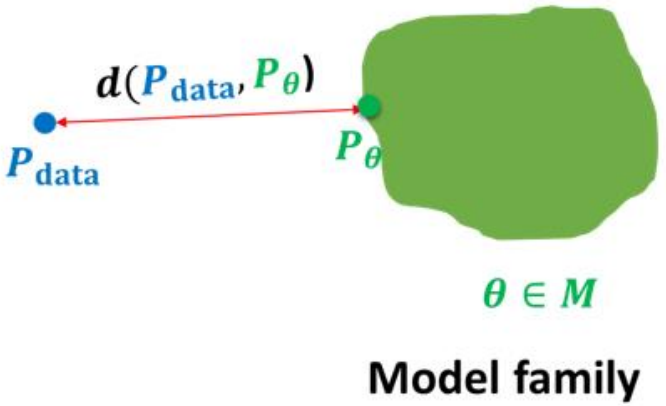
$$\theta^* = \arg\max_\theta \log \mathcal{L}(\theta, \mathcal{D})$$

Recap

Images of dogs



$x_i \sim P_{\text{data}}$
$i = 1, 2, \ldots, n$



Dog Image Generator

- Machine learning methods:
  - **Model Family**
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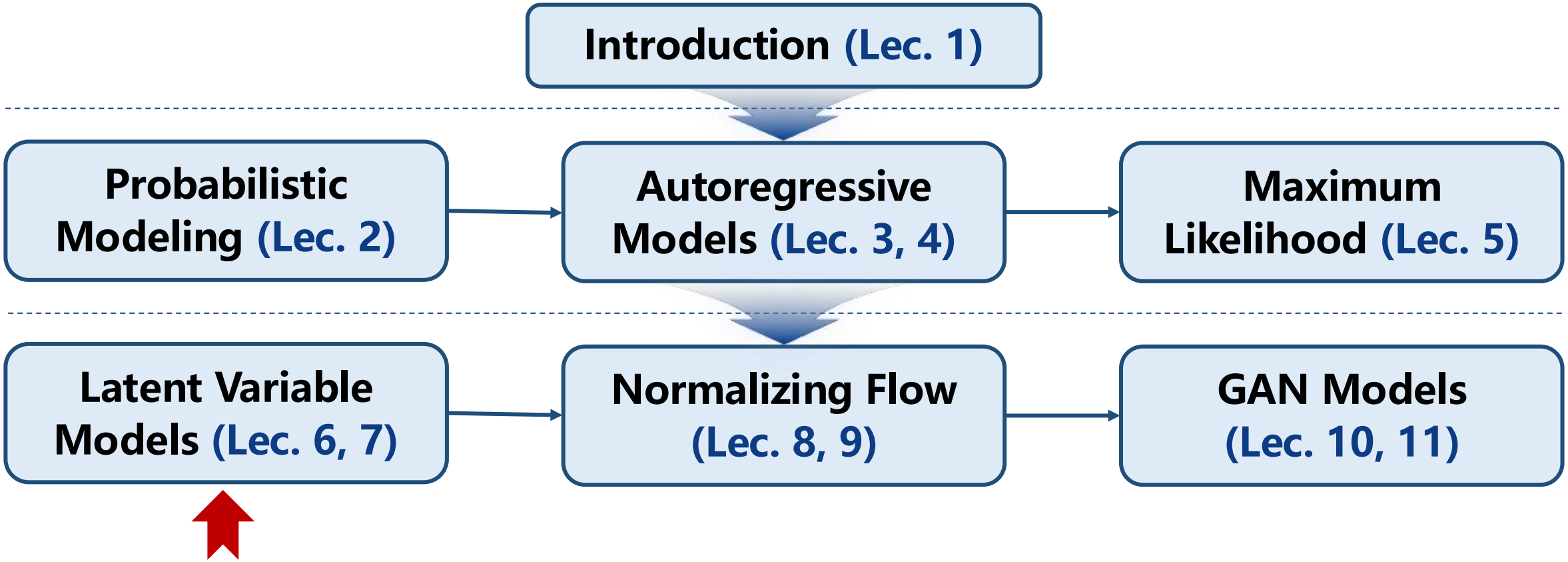
- Functions of generative models:
  - **Generation**
  - **Density Estimation (AR)**
  - **Representation Learning**

$$\theta^* = \arg\max_\theta \log \mathcal{L}(\theta, \mathcal{D})$$

Recap

- **Autoregressive Models**
  - Chain rule based factorization (fully general)
  - Compact representation via conditional independence / parameterization
- **Pros**
  - **Easy to evaluate likelihood**
  - Easy to train by maximum likelihood
- **Cons**
  - Requires an ordering
  - Generation is sequential
  - **Cannot learn features in an unsupervised manner**

Roadmap



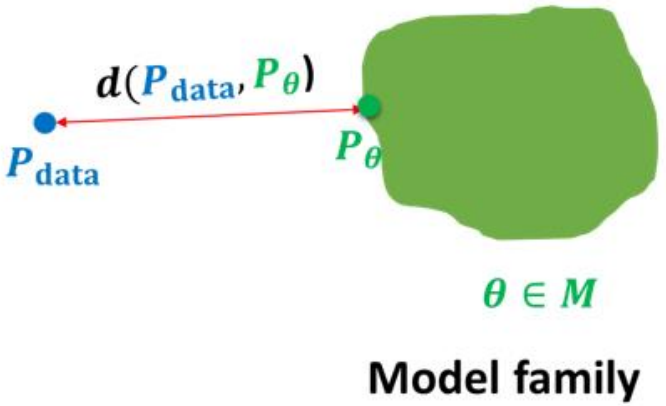
Introduction to various generative models

Recap

Images of dogs



$x_i \sim P_{\text{data}}$
$i = 1, 2, \ldots, n$



Dog Image Generator

- Machine learning methods:
  - **Model Family**
  - **Learning Objective**
  - **Optimization**
- Generative model functions:
  - **Generation**
  - **Density Estimation (AR)**
  - **Representation Learning**
- **Question: How to discover meaningful representations of x?**



Lecture Outline

- **Latent Variables Models**
- **Gaussian Mixture Models (GMM)**
- **Maximum Likelihood for Latent Variables Models**
- **Variational Inference**



Lecture Outline

- **Latent Variables Models**
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Latent Variable Models

- **Latent Variables**



Sample images from the CelebA dataset

Liu, Luo, Wang & Tang, Deep Learning Face Attributes in the Wild, ICCV 2015.

Latent Variable Models

- **Latent Variables**

Age

Gender

Hair color

With Eyeglasses

…



➢ Human faces can feature a lot of interesting **common characteristics** / **attributes** / **principal variations**

Latent Variable Models

- **Latent Variables**

Age

Gender

Hair color

With Eyeglasses

…



- Human faces can feature a lot of interesting **common characteristics**
- Given a state of these factors, a face image is **semantically determined**

Latent Variable Models

- **Latent Variables**

Age

Gender

Hair color

With Eyeglasses

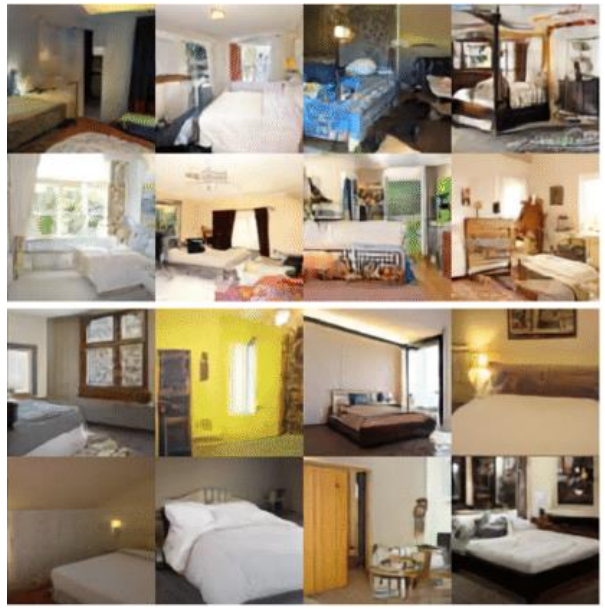
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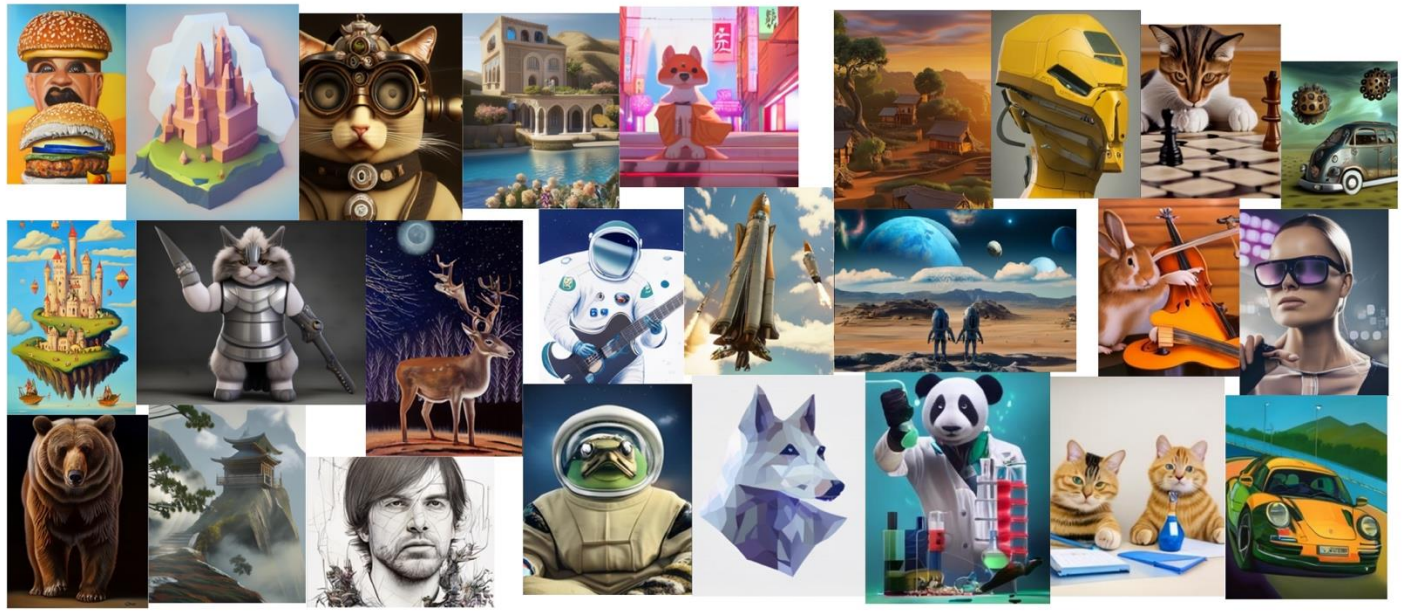
- ➢ These factors are **implicit (latent)** ➔ How to obtain given data samples?
- ➢ For face images, the facial characteristics can be manually labeled

Latent Variable Models

- **Latent Variables**



LSUN Bedroom Dataset

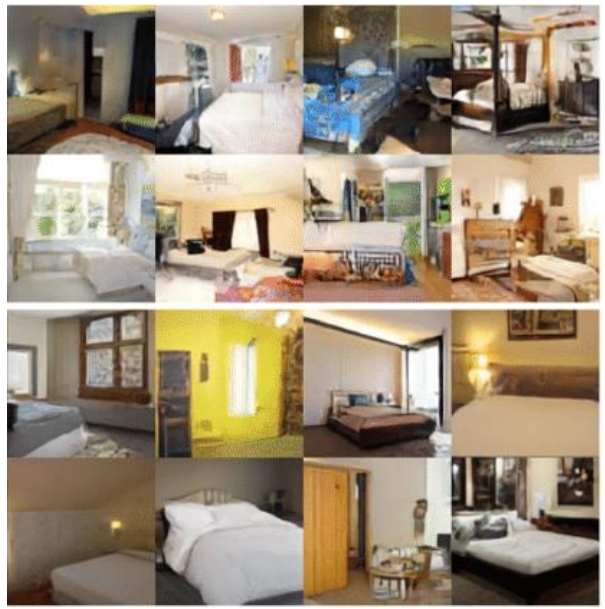


LAION Dataset

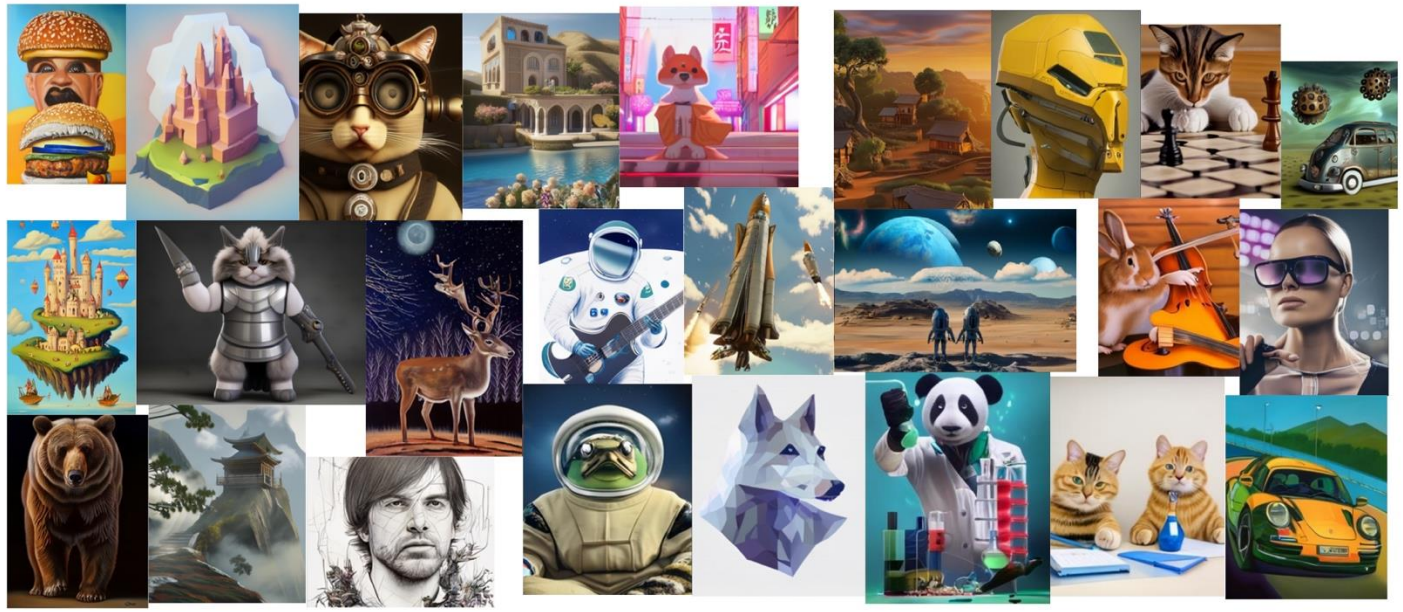
- These factors are **implicit (latent)** ➔ How to obtain given data samples?
- **Problem:** How to obtain the characteristics for **arbitrary dataset**?

Latent Variable Models

- **Latent Variables**



LSUN Bedroom Dataset



LAION Dataset

- **Challenge:** **Automatically** learn meaningful latent representations from x
- **Motivation:** **Explicitly model and learn** them using **latent variables z**

Latent Variable Models

- **Latent Variable Models**



➢ A latent variable model defines a probability distribution

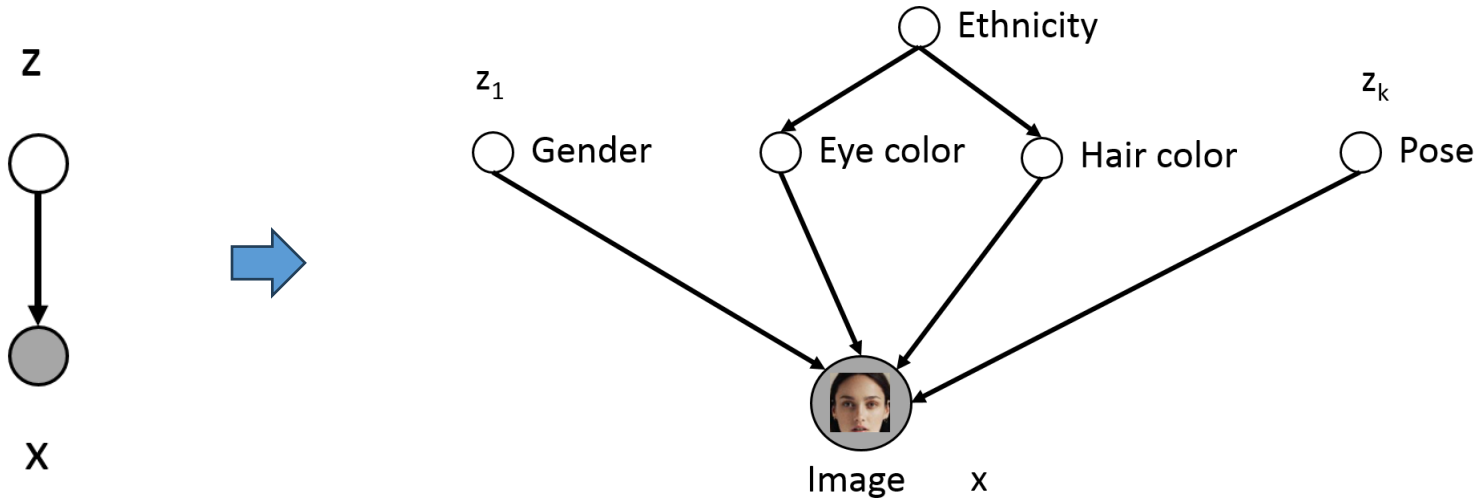
$$p(x, z) = p(x \mid z)\, p(z)$$

✓ **Observed variables x:** high-dimensional objects in the training set

✓ **Latent variables z: NOT explicitly provided**, but associated with x

Latent Variable Models

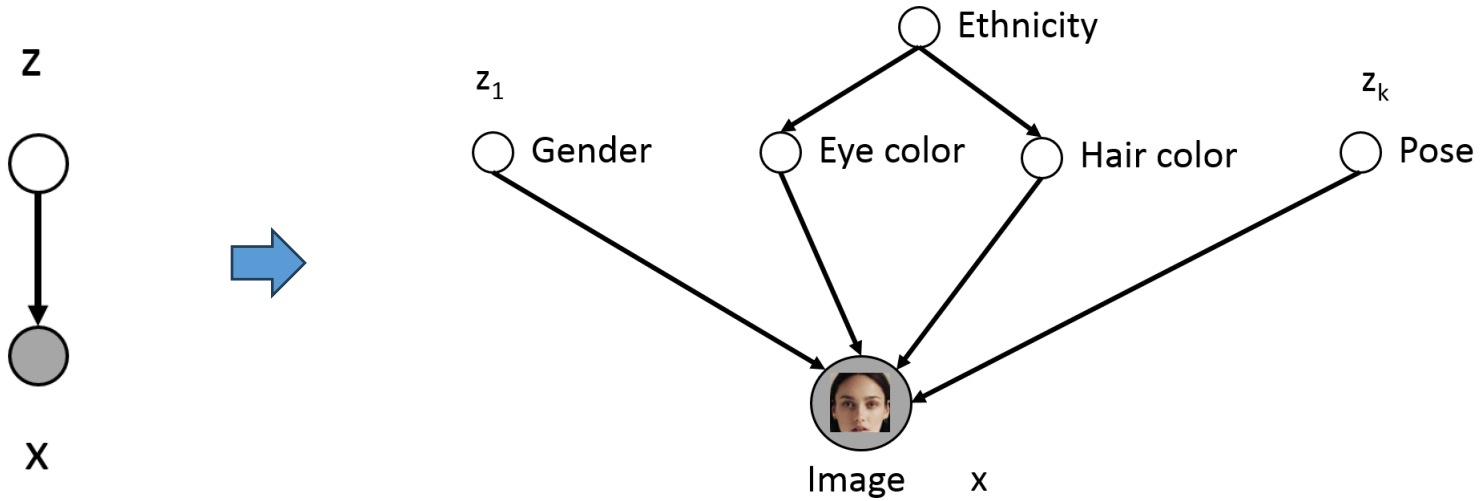
- **Latent Variable Models: The Face Image Example**



- **Observed variables x:** low-level pixel values
- **Latent variables z:** high-level features (e.g., gender, age, etc.)
  - ✓ If z is chosen properly, $p(x \mid z)$ could be **much simpler** than $p(x)$
  - ✓ If $p(x, z)$ is properly trained, we can **infer** the features via $p(z \mid x)$

Latent Variable Models

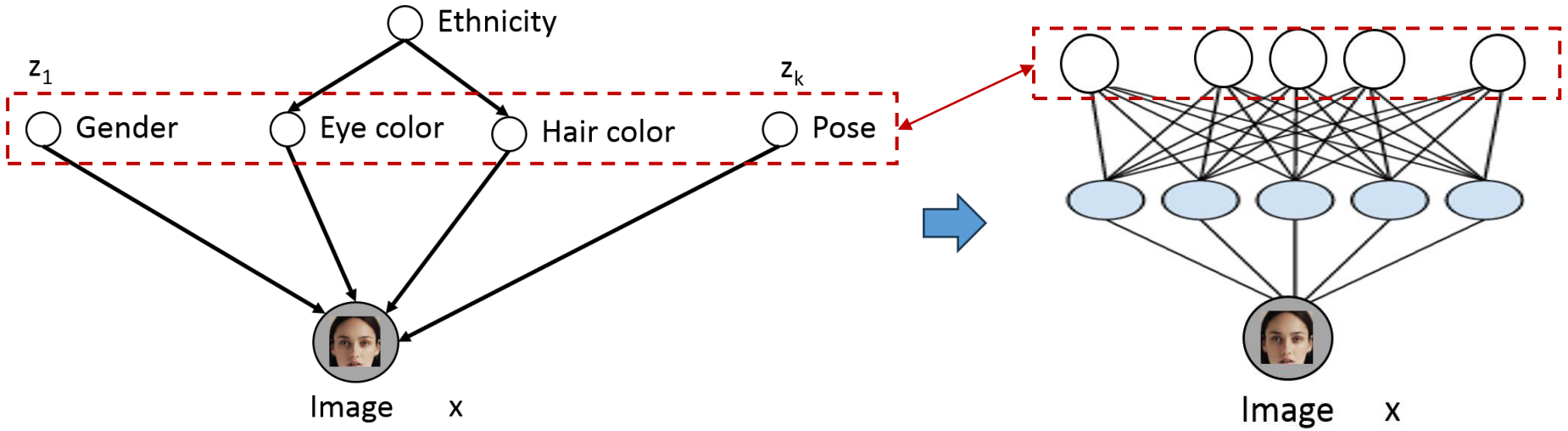
- **Latent Variable Models: The Face Image Example**



- **Observed variables x:** low-level pixel values
- **Latent variables z:** high-level features (e.g., gender, age, etc.)
  - ✓ $p(x \mid z)$: simpler modeling; $p(z \mid x)$: feature inference
- **Challenges:** **Very difficult to manually specify all representative z**

Latent Variable Models

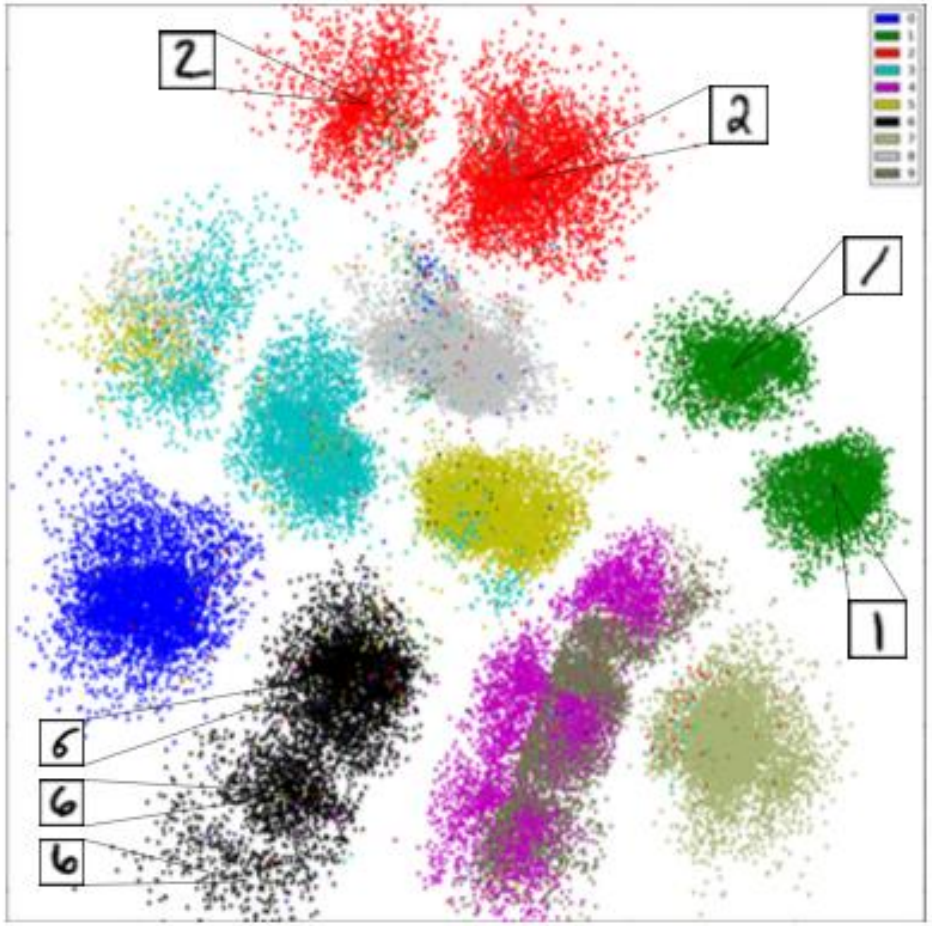
- **Latent Variable Models: The Face Image Example**



- $z \rightarrow x$ is **parameterized** by a neural network and **learned from data**
  - ✓ For example: $z \sim \mathcal{N}(0, I)$, $p(x \mid z) = \mathcal{N}\left(\mu_\theta(z), \Sigma_\theta(z)\right)$
- **Goal:** z corresponds to **meaningful latent factors of variation** (feature)
  - ✓ We want to learn a **structured** latent space Z (unsupervised)

Latent Variable Models

- **Structured Latent Space**



Unsupervised clustering of MNIST digits using deep latent variable models

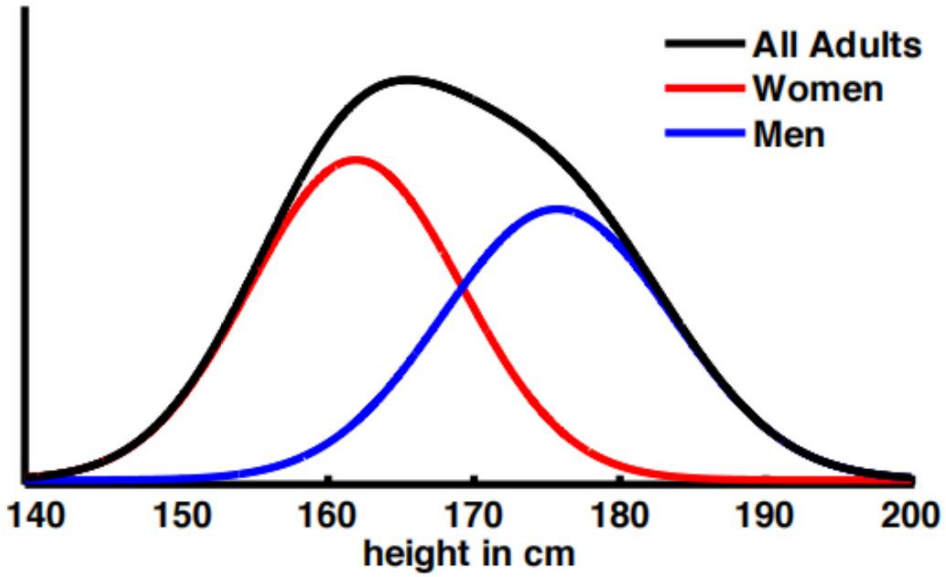


Lecture Outline

- **Latent Variables Models**
- **Gaussian Mixture Models (GMM)**
- **Maximum Likelihood for Latent Variables Models**
- **Variational Inference**

Gaussian Mixture Models (GMM)

- **A Toy Example: Height Estimation**
  - The overall height distribution is a mixture of two Gaussians



  - **Question:** How to estimate the distribution given only the height values?
    - ✓ 'Gender' is the latent variable, which determines data generation

Gaussian Mixture Models (GMM)

- **Components of GMM**
  - **Latent Variable ('prior')**

    $p(z) = \text{Categorical}(1, \ldots, K)$

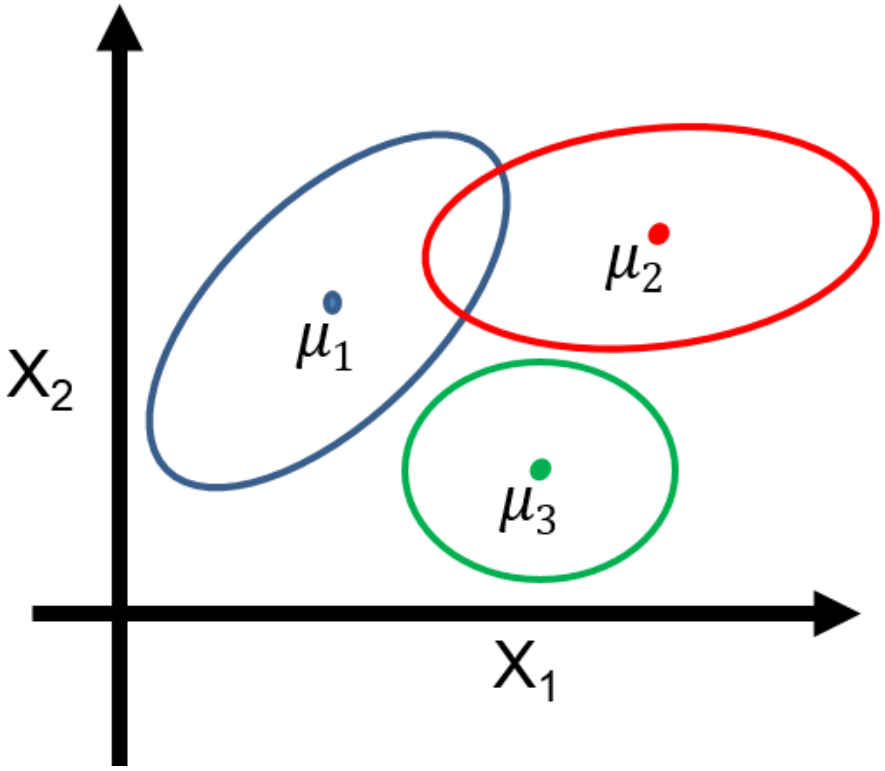
  - **Conditional Distribution**

    $p(x \mid z = k) = \mathcal{N}(\mu_k, \Sigma_k)$

  - **Generation Process**

    ① Pick a component $k \sim p(z)$

    ② Sample a data point from the corresponding Gaussian

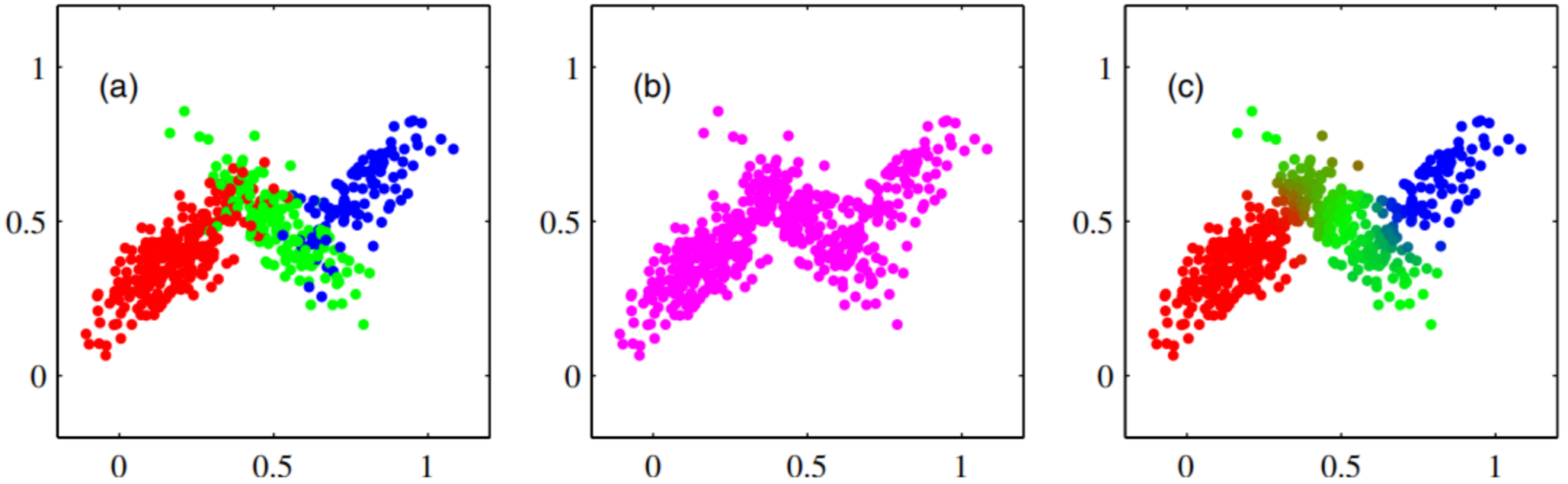


<height, weight> pairs of 3 kinds of animals

Gaussian Mixture Models (GMM)

- **Mixture of Gaussians**

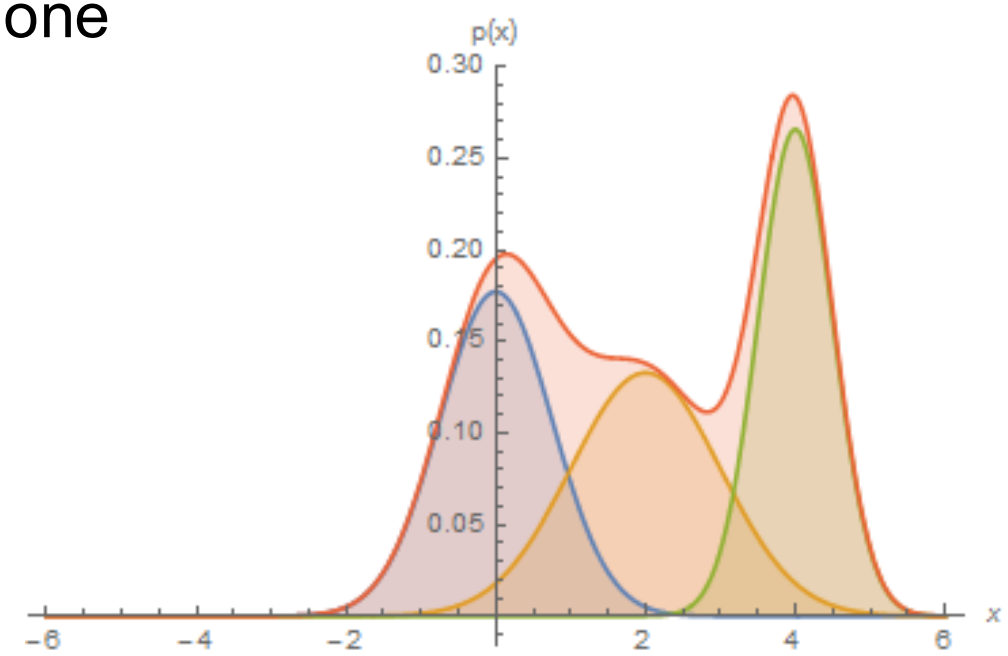
Mixture of 3 components Unlabeled Samples x $p(z \mid x)$



- **Clustering:** The posterior $p(z \mid x)$ identifies the mixture component
- **Representation:** Learn latent components $p(z \mid x)$ from **unlabeled data**

Gaussian Mixture Models (GMM)

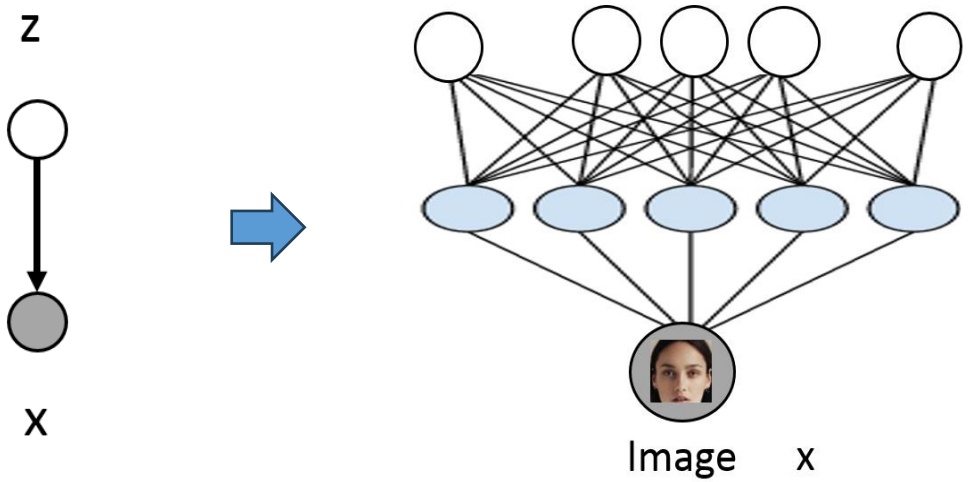
- **Mixture of Models**
  - **Alternative Motivation: Combine simple models** into a more complex and expressive one



$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{x}, \mathbf{z}) = \sum_{\mathbf{z}} p(\mathbf{z}) p(\mathbf{x} \mid \mathbf{z}) = \sum_{k=1}^{K} p(\mathbf{z} = k) \underbrace{\mathcal{N}(\mathbf{x}; \mu_k, \Sigma_k)}_{\text{component}}$$

Gaussian Mixture Models (GMM)

- **Mixture of Models**
  - **Motivation:** Push the mixture to an **infinite number of Gaussians**



  - ✓ $\boldsymbol{z} \sim \mathcal{N}(0, \boldsymbol{I})$
  - ✓ $p(\boldsymbol{x} \mid \boldsymbol{z}) = \mathcal{N}\left(\mu_\theta(\boldsymbol{z}), \Sigma_\theta(\boldsymbol{z})\right)$, where μ_θ and Σ_θ are neural networks
  - Even though $\boldsymbol{p(x \mid z)}$ **is simple**, $\boldsymbol{p(x)}$ is very **flexible and complex**

Summary

- **Representation Learning**
  - Latent variables z ➔ **Implicit / Hidden structure of data distribution**
  - Naturally suitable for unsupervised learning (e.g., clustering, etc.)
- **Distribution Factorization**
  - Define complex models $p(x)$ with simpler building blocks $p(x \mid z)$
- **"No Free Lunch"**
  - **The log-likelihood is hard to maximize** (compared to AR models)

$$p(x) = \sum_z p(z)\ p(x|z) \qquad p(x) = \int p(z)\ p(x|z)\, dz$$



Content

- **Latent Variables Models**
- **Gaussian Mixture Models (GMM)**
- **Maximum Likelihood for Latent Variables Models**
- **Variational Inference**

Maximum Likelihood for Latent Variables Models

- **Marginal Likelihood: Discrete Example**
  - Suppose we have a model for the joint distribution: $p(X, Z; \theta)$
    - ✓ X: observed variables
    - ✓ Z: unobserved variables
  - The probability / likelihood of observing a training data point $\bar{x}$?

$$p(\mathbf{X} = \bar{\mathbf{x}}; \theta)$$



$$\sum_{\mathbf{z}} p(\mathbf{X} = \bar{\mathbf{x}}, \mathbf{Z} = \mathbf{z}; \theta) = \sum_{\mathbf{z}} p(\bar{\mathbf{x}}, \mathbf{z}; \theta)$$

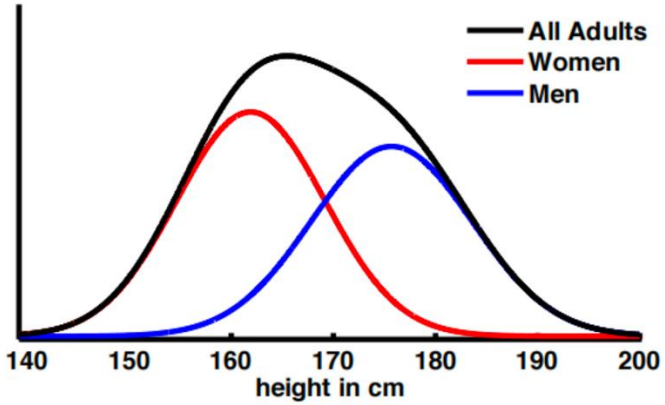
➔ **Summing over all possible latent z that could generate $\bar{x}$**

Maximum Likelihood for Latent Variables Models

- **Marginal Likelihood: Discrete Example**
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$$p(\mathbf{X} = \bar{\mathbf{x}}; \theta)$$

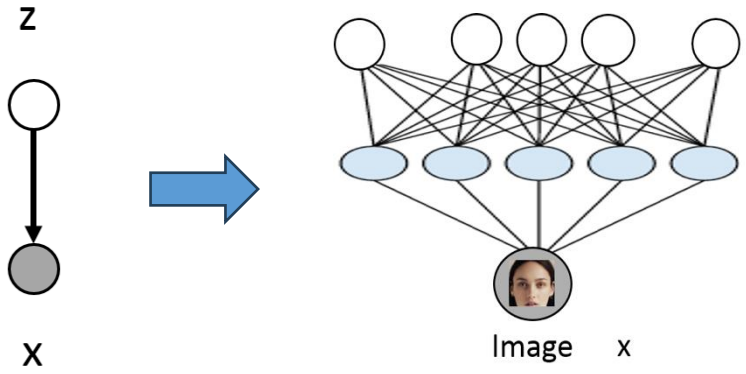
$$\sum_{\mathbf{z}} p(\mathbf{X} = \bar{\mathbf{x}}, \mathbf{Z} = \mathbf{z}; \theta) = \sum_{\mathbf{z}} p(\bar{\mathbf{x}}, \mathbf{z}; \theta)$$



➔ **Summing over all possible latent z that could generate $\bar{x}$**

Maximum Likelihood for Latent Variables Models

- **Marginal Likelihood: Continuous Example**



➢ A mixture of an infinite number of Gaussians

✓ $z \sim \mathcal{N}(0, I)$, unobserved at train time

✓ $p(x \mid z) = \mathcal{N}\left(\mu_\theta(z), \Sigma_\theta(z)\right)$, where μ_θ and Σ_θ are neural networks

➢ The probability / likelihood of observing a training data point $\bar{x}$?

$$\int_{\mathbf{z}} p(\mathbf{X} = \bar{\mathbf{x}}, \mathbf{Z} = \mathbf{z}; \theta) d\mathbf{z} = \int_{\mathbf{z}} p(\bar{\mathbf{x}}, \mathbf{z}; \theta) d\mathbf{z}$$



Maximum Likelihood for Latent Variables Models

- **Difficulty of Marginal Likelihood Estimation**
  - Given dataset $\mathcal{D} = \{x^{(1)}, \ldots, x^{(m)}\}$, x are observed variables (pixel values), z are unobserved variables (cluster, class ID, etc.)
  - **Question:** What if we still want to train via maximum likelihood?

$$\sum_{\mathbf{z}} p(\mathbf{X} = \bar{\mathbf{x}}, \mathbf{Z} = \mathbf{z}; \theta) = \sum_{\mathbf{z}} p(\bar{\mathbf{x}}, \mathbf{z}; \theta)$$

$$\int_{\mathbf{z}} p(\mathbf{X} = \bar{\mathbf{x}}, \mathbf{Z} = \mathbf{z}; \theta) d\mathbf{z} = \int_{\mathbf{z}} p(\bar{\mathbf{x}}, \mathbf{z}; \theta) d\mathbf{z}$$

① Write out the (log-) likelihood expression $\mathcal{L}(\theta; \mathcal{D})$

② Find the optimal θ maximizing $\mathcal{L}(\theta; \mathcal{D})$

Maximum Likelihood for Latent Variables Models

- **Difficulty of Marginal Likelihood Estimation**
  - Given dataset $\mathcal{D} = \{x^{(1)}, \ldots, x^{(m)}\}$, x are observed variables (pixel values), z are unobserved variables (cluster, class ID, etc.)
  - The log-likelihood is

$$\log \prod_{\mathbf{x} \in \mathcal{D}} p(\mathbf{x}; \theta) = \sum_{\mathbf{x} \in \mathcal{D}} \log p(\mathbf{x}; \theta) = \sum_{\mathbf{x} \in \mathcal{D}} \log \sum_{\mathbf{z}} p(\mathbf{x}, \mathbf{z}; \theta) \quad \Big| \quad \int_{\mathbf{z}} p(\mathbf{x}, \mathbf{z}; \theta) d\mathbf{z}$$

  - $\log \sum_{\mathbf{z}} p(\mathbf{x}, \mathbf{z}; \theta)$ and $\int_{\mathbf{z}} p(\mathbf{x}, \mathbf{z}; \theta) d\mathbf{z}$ can be **intractable to evaluate**
    - ✓ $\log \sum_{\mathbf{z}} p(\mathbf{x}, \mathbf{z}; \theta)$: Summation of all z values is expensive to compute
    - ✓ $\int_{\mathbf{z}} p(\mathbf{x}, \mathbf{z}; \theta) d\mathbf{z}$: No close-form, hard to maximize

Approximation needed!

Maximum Likelihood for Latent Variables Models

- **Approximation ① : Naïve Monte Carlo**
  - For discrete $z \in \mathcal{Z}$, the likelihood function can be written as

$$p_\theta(\mathbf{x}) = \sum_{\text{All values of } \mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) = |\mathcal{Z}| \sum_{\mathbf{z} \in \mathcal{Z}} \frac{1}{|\mathcal{Z}|} p_\theta(\mathbf{x}, \mathbf{z}) = |\mathcal{Z}| \mathbb{E}_{\mathbf{z} \sim Uniform(\mathcal{Z})} [p_\theta(\mathbf{x}, \mathbf{z})]$$

  - By Monte Carlo Estimation

    ① Sample $z^{(1)}, \ldots, z^{(k)}$ **uniformly at random**

    ② Approximate expectation with sample average

$$\sum_{\mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) \approx |\mathcal{Z}| \frac{1}{k} \sum_{j=1}^{k} p_\theta(\mathbf{x}, \mathbf{z}^{(j)})$$

Maximum Likelihood for Latent Variables Models

- **Approximation ① : Naïve Monte Carlo**

  ① Sample $z^{(1)}, \ldots, z^{(k)}$ **uniformly at random**

  ② Approximate expectation with sample average

$$\sum_{\mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) \approx |\mathcal{Z}| \frac{1}{k} \sum_{j=1}^{k} p_\theta(\mathbf{x}, \mathbf{z}^{(j)})$$

Does it work well? If not, why?

Maximum Likelihood for Latent Variables Models

- **Approximation ① : Naïve Monte Carlo**

  ① Sample $z^{(1)}, \ldots, z^{(k)}$ **uniformly at random**

  ② Approximate expectation with sample average

$$\sum_{\mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) \approx |\mathcal{Z}| \frac{1}{k} \sum_{j=1}^{k} p_\theta(\mathbf{x}, \mathbf{z}^{(j)})$$

  ➢ **Works in theory, but not in practice!**

  ✓ For most $z^{(j)}$, $p_\theta\left(x, z^{(j)}\right)$ is very low

  ✓ For $z^{(j)}$ with large $p_\theta\left(x, z^{(j)}\right)$, uniform sampling may never "hit"

  ➔ We need **a clever way to select $z^{(j)}$** (based on data samples)

Maximum Likelihood for Latent Variables Models

- **Approximation ②: Importance Sampling**
  - For discrete $z \in \mathcal{Z}$, the likelihood function can be written as

$$p_\theta(\mathbf{x}) = \sum_{\text{All values of } \mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) = |\mathcal{Z}| \sum_{\mathbf{z} \in \mathcal{Z}} \frac{1}{|\mathcal{Z}|} p_\theta(\mathbf{x}, \mathbf{z}) = |\mathcal{Z}| \mathbb{E}_{\mathbf{z} \sim Uniform(\mathcal{Z})} [p_\theta(\mathbf{x}, \mathbf{z})]$$

Uniform Sampling

$$p_\theta(\mathbf{x}) = \sum_{\text{All values of } \mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) = \sum_{\mathbf{z} \in \mathcal{Z}} \frac{q(\mathbf{z})}{q(\mathbf{z})} p_\theta(\mathbf{x}, \mathbf{z}) = \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right]$$

Sampling by arbitrary $q(z)$

  - By Monte Carlo Estimation
    - ① Sample $z^{(1)}, \ldots, z^{(k)}$ according to $q(z)$
    - ② Approximate expectation with sample average $p_\theta(\mathbf{x}) \approx \frac{1}{k} \sum_{j=1}^{k} \frac{p_\theta(\mathbf{x}, \mathbf{z}^{(j)})}{q(\mathbf{z}^{(j)})}$

Maximum Likelihood for Latent Variables Models

- **Approximation ②: Importance Sampling**

$$p_\theta(\mathbf{x}) = \sum_{\text{All values of } \mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) = \sum_{\mathbf{z} \in \mathcal{Z}} \frac{q(\mathbf{z})}{q(\mathbf{z})} p_\theta(\mathbf{x}, \mathbf{z}) = \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right]$$

$$\Rightarrow \quad p_\theta(\mathbf{x}) \approx \frac{1}{k} \sum_{j=1}^{k} \frac{p_\theta(\mathbf{x}, \mathbf{z}^{(j)})}{q(\mathbf{z}^{(j)})}$$

Sampling by arbitrary $q(z)$

  - **Question:** What is a good choice for $q(z)$?
    - ✓ Intuitively, $q(z)$ should place most of its mass on z **where $p_\theta(x, z)$ is large for the observed x**

      ➔ Frequently sample those z that **"explain" the observed x well**

Introducing $q(z)$ ➔ Making $p_\theta(x)$ tractable and thus can be optimized!

Maximum Likelihood for Latent Variables Models

- **Evidence Lower Bound (ELBO)**
  - The marginal log-likelihood can now be written as

$$\log\left(\sum_{\mathbf{z}\in\mathcal{Z}} p_\theta(\mathbf{x},\mathbf{z})\right) = \log\left(\sum_{\mathbf{z}\in\mathcal{Z}} \frac{q(\mathbf{z})}{q(\mathbf{z})} p_\theta(\mathbf{x},\mathbf{z})\right) = \log\left(\mathbb{E}_{\mathbf{z}\sim q(\mathbf{z})}\left[\frac{p_\theta(\mathbf{x},\mathbf{z})}{q(\mathbf{z})}\right]\right)$$

  - We aim to approximate $\mathbb{E}_{z\sim q(z)}$ by Monte Carlo, but it is within $\log(\cdot)$
    - ✓ And we cannot directly flip $\mathbb{E}_{z\sim q(z)}$ and $\log(\cdot)$

$$\mathbb{E}_{\mathbf{z}\sim q(\mathbf{z})}\left[\log\left(\frac{p_\theta(\mathbf{x},\mathbf{z})}{q(\mathbf{z})}\right)\right] \neq \log\left(\mathbb{E}_{\mathbf{z}\sim q(\mathbf{z})}\left[\frac{p_\theta(\mathbf{x},\mathbf{z})}{q(\mathbf{z})}\right]\right)$$

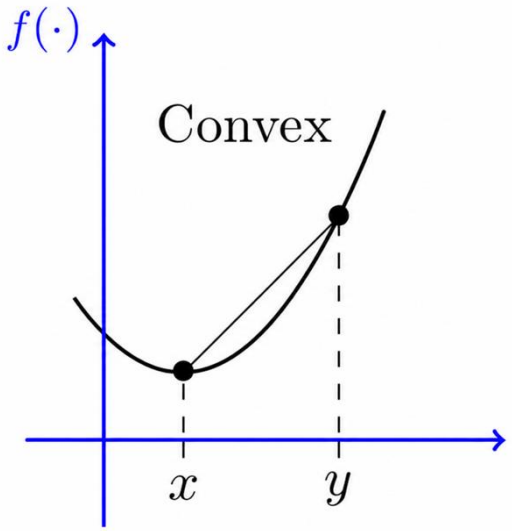
    - ✓ Therefore, we need an alternative approach to expose $\mathbb{E}_{z\sim q(z)}$

Maximum Likelihood for Latent Variables Models

- **Evidence Lower Bound (ELBO)**
  - The marginal log-likelihood can now be written as

$$\log\left(\sum_{\mathbf{z}\in\mathcal{Z}} p_\theta(\mathbf{x},\mathbf{z})\right) = \log\left(\sum_{\mathbf{z}\in\mathcal{Z}} \frac{q(\mathbf{z})}{q(\mathbf{z})} p_\theta(\mathbf{x},\mathbf{z})\right) = \log\left(\mathbb{E}_{\mathbf{z}\sim q(\mathbf{z})}\left[\frac{p_\theta(\mathbf{x},\mathbf{z})}{q(\mathbf{z})}\right]\right)$$

  - Using **Jensen Inequality**

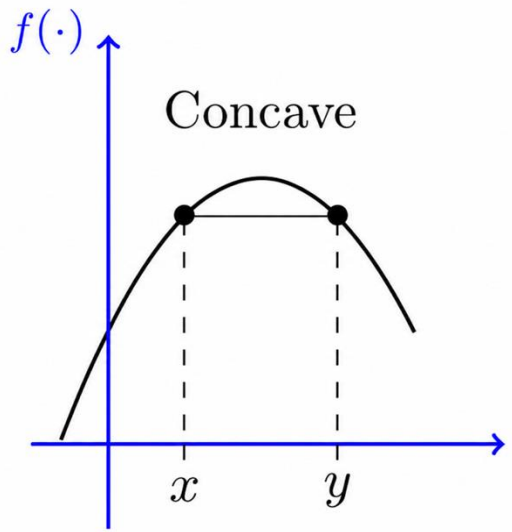


$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

If $f(\cdot)$ is convex

$$f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$$

If $f(\cdot)$ is concave



Maximum Likelihood for Latent Variables Models

- **Evidence Lower Bound (ELBO)**
  - The marginal log-likelihood can now be written as

$$\log\left(\sum_{\mathbf{z}\in\mathcal{Z}} p_\theta(\mathbf{x},\mathbf{z})\right) = \log\left(\sum_{\mathbf{z}\in\mathcal{Z}} \frac{q(\mathbf{z})}{q(\mathbf{z})} p_\theta(\mathbf{x},\mathbf{z})\right) = \log\left(\mathbb{E}_{\mathbf{z}\sim q(\mathbf{z})}\left[\frac{p_\theta(\mathbf{x},\mathbf{z})}{q(\mathbf{z})}\right]\right)$$

  - Using **Jensen Inequality**

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)] \qquad f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$$

If $f(\cdot)$ is convex — If $f(\cdot)$ is concave

$$\log\left(\sum_{\mathbf{z}\in\mathcal{Z}} p_\theta(\mathbf{x},\mathbf{z})\right) = \log\left(\mathbb{E}_{\mathbf{z}\sim q(\mathbf{z})}\left[\frac{p_\theta(\mathbf{x},\mathbf{z})}{q(\mathbf{z})}\right]\right) \geq \mathbb{E}_{\mathbf{z}\sim q(\mathbf{z})}\left[\log\left(\frac{p_\theta(\mathbf{x},\mathbf{z})}{q(\mathbf{z})}\right)\right]$$

Marginal Log-likelihood — **Evidence Lower Bound (ELBO)**

Summary

- **Motivation:** Explicitly model the latent variable z via $p_\theta(x, z)$
  - **Data sample x:** Observed
  - **Latent variable z:** Unobserved, but "explains" the generation of x

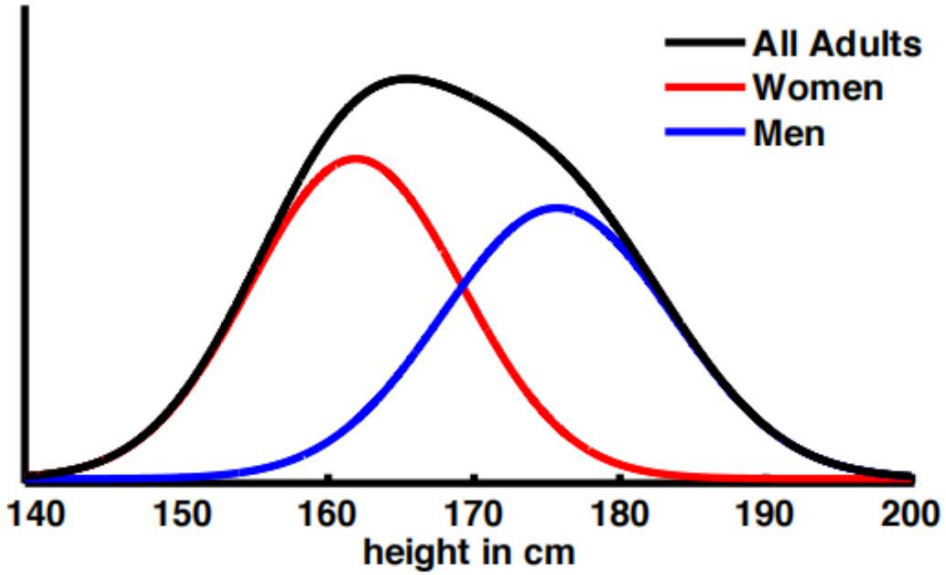


$$p(x, z) = p(x \mid z)\, p(z)$$

Summary

- **Motivation:** Explicitly model the latent variable z via $p_\theta(x, z)$
  - **Data sample x:** Observed
  - **Latent variable z:** Unobserved, but "explains" the generation of x





Height values (x) are observed, Gender (z) are latent (but explains x)

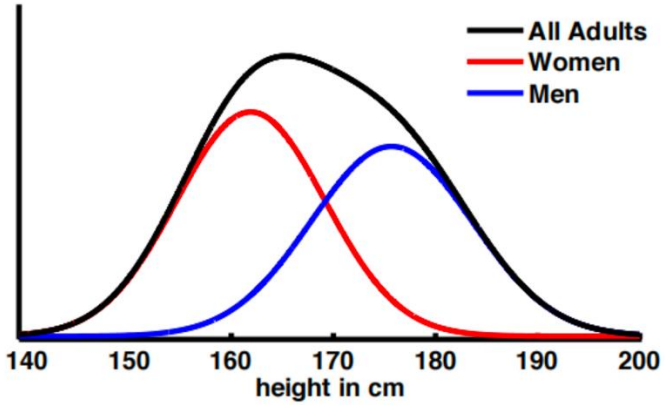
Summary

- **Motivation:** Explicitly model the latent variable z via $p_\theta(x, z)$
  - **Data sample x:** Observed
  - **Latent variable z:** Unobserved, but "explains" the generation of x
- **Learning $p_\theta(x, z)$ via Maximum Likelihood**
  - The Likelihood of a data sample $\bar{x}$: $p(\mathbf{X} = \bar{\mathbf{x}}; \theta)$



$$p(\mathbf{X} = \bar{\mathbf{x}}; \theta)$$

$$\sum_{\mathbf{z}} p(\mathbf{X} = \bar{\mathbf{x}}, \mathbf{Z} = \mathbf{z}; \theta) = \sum_{\mathbf{z}} p(\bar{\mathbf{x}}, \mathbf{z}; \theta)$$



Summary

- **Motivation:** Explicitly model the latent variable z via $p_\theta(x, z)$
  - **Data sample x:** Observed
  - **Latent variable z:** Unobserved, but "explains" the generation of x
- **Learning $p_\theta(x, z)$ via Maximum Likelihood**
  - The Likelihood of a data sample $\bar{x}$: $p(\mathbf{X} = \bar{\mathbf{x}}; \theta)$
    - ✓ **Intractable** in practice: $p_\theta(\mathbf{x}) = \sum_{\text{All values of } \mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z})$
    - ✓ Approximation ①: Naïve Monte Carlo

$$p_\theta(\mathbf{x}) = \sum_{\text{All values of } \mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) = |\mathcal{Z}| \sum_{\mathbf{z} \in \mathcal{Z}} \frac{1}{|\mathcal{Z}|} p_\theta(\mathbf{x}, \mathbf{z}) = |\mathcal{Z}| \mathbb{E}_{\mathbf{z} \sim Uniform(\mathcal{Z})} [p_\theta(\mathbf{x}, \mathbf{z})]$$



Summary

- **Motivation:** Explicitly model the latent variable z via $p_\theta(x, z)$
  - **Data sample x:** Observed
  - **Latent variable z:** Unobserved, but "explains" the generation of x



- **Learning $p_\theta(x, z)$ via Maximum Likelihood**
  - The Likelihood of a data sample $\bar{x}$: $p(\mathbf{X} = \bar{\mathbf{x}}; \theta)$
    - ✓ **Intractable** in practice: $p_\theta(\mathbf{x}) = \sum_{\text{All values of } \mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z})$
    - ✓ Approximation ①: Naïve Monte Carlo
    - ✓ Approximation ②: **Importance Sampling**

$$p_\theta(\mathbf{x}) = \sum_{\text{All values of } \mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) = \sum_{\mathbf{z} \in \mathcal{Z}} \frac{q(\mathbf{z})}{q(\mathbf{z})} p_\theta(\mathbf{x}, \mathbf{z}) = \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right]$$

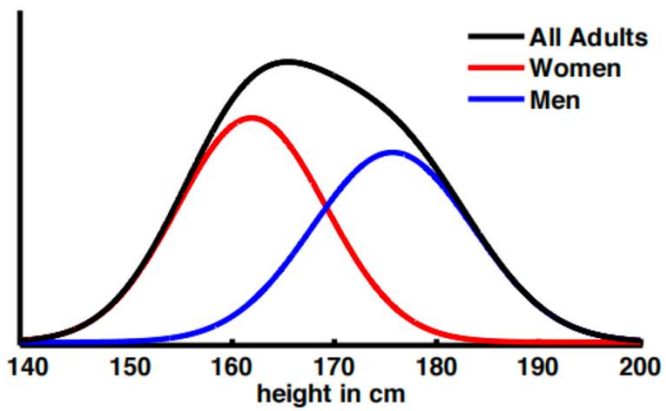
Sampling by arbitrary $q(z)$

Summary

- **Motivation:** Explicitly model the latent variable z via $p_\theta(x, z)$
  - **Data sample x:** Observed
  - **Latent variable z:** Unobserved, but "explains" the generation of x
- **Learning $p_\theta(x, z)$ via Maximum Likelihood**
  - The Likelihood of a data sample $\bar{x}$: $p(\mathbf{X} = \bar{\mathbf{x}}; \theta)$
    - ✓ **Intractable** in practice: $p_\theta(\mathbf{x}) = \sum_{\text{All values of } \mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z})$
    - ✓ Approximation ①: Naïve Monte Carlo
    - ✓ Approximation ②: **Importance Sampling**



$$p(\mathbf{X} = \bar{\mathbf{x}}; \theta) \Rightarrow \sum_{\mathbf{z}} p(\mathbf{X} = \bar{\mathbf{x}}, \mathbf{Z} = \mathbf{z}; \theta) = \sum_{\mathbf{z}} p(\bar{\mathbf{x}}, \mathbf{z}; \theta)$$



Summary

- **Learning $p_\theta(x, z)$ via Maximum Likelihood**

  ➢ Use **importance sampling** for likelihood approximation

$$p_\theta(\mathbf{x}) = \sum_{\text{All values of } \mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) = \sum_{\mathbf{z} \in \mathcal{Z}} \frac{q(\mathbf{z})}{q(\mathbf{z})} p_\theta(\mathbf{x}, \mathbf{z}) = \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right]$$

$$\Rightarrow \quad p_\theta(\mathbf{x}) \approx \frac{1}{k} \sum_{j=1}^{k} \frac{p_\theta(\mathbf{x}, \mathbf{z}^{(j)})}{q(\mathbf{z}^{(j)})}$$

Sampling by arbitrary $q(z)$



  ➢ For log-likelihood per sample

$$\log \left(\sum_{\mathbf{z} \in \mathcal{Z}} p_\theta(\mathbf{x}, \mathbf{z}) \right) = \log \left(\sum_{\mathbf{z} \in \mathcal{Z}} \frac{q(\mathbf{z})}{q(\mathbf{z})} p_\theta(\mathbf{x}, \mathbf{z}) \right) = \log \left(\mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right] \right)$$

  ✓ Cannot directly apply Monte Carlo Estimation

  ✓ Use **Jensen's Inequality** for switching $\mathbb{E}_{z \sim q(z)}$ and $\log(\cdot)$

Summary

- **Learning $p_\theta(x, z)$ via Maximum Likelihood**
  - Use **importance sampling** for likelihood approximation

$$p_\theta(\mathbf{x}) = \sum_{\text{All values of } \mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) = \sum_{\mathbf{z} \in \mathcal{Z}} \frac{q(\mathbf{z})}{q(\mathbf{z})} p_\theta(\mathbf{x}, \mathbf{z}) = \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right]$$



  - Use **Jensen's Inequality** for deriving ELBO

$$\log \left(\sum_{\mathbf{z} \in \mathcal{Z}} p_\theta(\mathbf{x}, \mathbf{z}) \right) = \log \left(\sum_{\mathbf{z} \in \mathcal{Z}} \frac{q(\mathbf{z})}{q(\mathbf{z})} p_\theta(\mathbf{x}, \mathbf{z}) \right) = \log \left(\mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right] \right)$$

$$f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$$

$$\log \left(\sum_{\mathbf{z} \in \mathcal{Z}} p_\theta(\mathbf{x}, \mathbf{z}) \right) = \log \left(\mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right] \right) \geq \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\log \left(\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right) \right]$$

Marginal Log-likelihood **Evidence Lower Bound (ELBO)**

Summary

- **Motivation:** Explicitly model the latent variable z via $p_\theta(x, z)$
  - **Data sample x:** Observed
  - **Latent variable z:** Unobserved, but "explains" the generation of x
- **Learning $p_\theta(x, z)$ via Maximum Likelihood**
  - Maximum likelihood ➔ Maximizing ELBO (*surrogate* objective)



Importance Sampling **Jensen's Inequality**

$$\log\left(\sum_{\mathbf{z}\in\mathcal{Z}} p_\theta(\mathbf{x},\mathbf{z})\right) = \log\left(\mathbb{E}_{\mathbf{z}\sim q(\mathbf{z})}\left[\frac{p_\theta(\mathbf{x},\mathbf{z})}{q(\mathbf{z})}\right]\right) \geq \mathbb{E}_{\mathbf{z}\sim q(\mathbf{z})}\left[\log\left(\frac{p_\theta(\mathbf{x},\mathbf{z})}{q(\mathbf{z})}\right)\right]$$

Marginal Log-likelihood **Evidence Lower Bound (ELBO)**



Lecture Outline

- **Latent Variables Models**
- **Gaussian Mixture Models (GMM)**
- **Maximum Likelihood for Latent Variables Models**
- **Variational Inference**

Variational Inference

- **Variational Inference (变分推断)**
  - For importance sampling, $q(z)$ is proposed to help estimate likelihood

$$p_\theta(\mathbf{x}) = \sum_{\text{All values of } \mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) = \sum_{\mathbf{z} \in \mathcal{Z}} \frac{q(\mathbf{z})}{q(\mathbf{z})} p_\theta(\mathbf{x}, \mathbf{z}) = \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right]$$

  - For **variational inference**, $q(z)$ is **something to be optimized**

$$\log \left(\sum_{\mathbf{z} \in \mathcal{Z}} p_\theta(\mathbf{x}, \mathbf{z}) \right) = \log \left(\mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right] \right) \geq \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[\log \left(\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \right) \right]$$

$$\log p_\theta(x) \geq \mathcal{L}(\theta, q)$$

Evidence Lower Bound (ELBO)

Variational Inference

- **Variational Inference and ELBO**
  - The Evidence Lower Bound (ELBO) holds for any $q(z)$

Marginal Log-likelihood

$$\log p(\mathbf{x};\theta) \geq \sum_{\mathbf{z}} q(\mathbf{z}) \log \left(\frac{p_\theta(\mathbf{x},\mathbf{z})}{q(\mathbf{z})} \right)$$

Evidence Lower Bound (ELBO)

$$= \sum_{\mathbf{z}} q(\mathbf{z}) \log p_\theta(\mathbf{x},\mathbf{z}) \underbrace{- \sum_{\mathbf{z}} q(\mathbf{z}) \log q(\mathbf{z})}_{\text{Entropy } H(q) \text{ of } q}$$

$$= \sum_{\mathbf{z}} q(\mathbf{z}) \log p_\theta(\mathbf{x},\mathbf{z}) + H(q)$$

Pushes $q(z)$ to put mass on z where $p_\theta(x,z)$ is large

Regularization Avoid overfitting on x

Variational Inference

- **Variational Inference and ELBO**
  - The Evidence Lower Bound (ELBO) holds for any $q(z)$

Marginal Log-likelihood

$$\log p(\mathbf{x};\theta) \geq \sum_{\mathbf{z}} q(\mathbf{z}) \log \left(\frac{p_\theta(\mathbf{x},\mathbf{z})}{q(\mathbf{z})} \right)$$

Evidence Lower Bound (ELBO)

$$= \sum_{\mathbf{z}} q(\mathbf{z}) \log p_\theta(\mathbf{x},\mathbf{z}) \underbrace{- \sum_{\mathbf{z}} q(\mathbf{z}) \log q(\mathbf{z})}_{\text{Entropy } H(q) \text{ of } q}$$

$$= \sum_{\mathbf{z}} q(\mathbf{z}) \log p_\theta(\mathbf{x},\mathbf{z}) + H(q)$$

  - Equality holds if $q(z) = p(z \mid x;\theta)$

$$\log p(\mathbf{x};\theta) = \sum_{\mathbf{z}} q(\mathbf{z}) \log p(\mathbf{z},\mathbf{x};\theta) + H(q)$$

***E-step of the EM algorithm**

Variational Inference

- **Variational Inference and ELBO**
  - A quick check on the bound when $q(z) = p(z \mid x; \theta)$

$$\log p(\mathbf{x};\theta) \geq \sum_{\mathbf{z}} q(\mathbf{z}) \log \left(\frac{p_\theta(\mathbf{x},\mathbf{z})}{q(\mathbf{z})} \right) \quad \Leftarrow \quad q(\mathbf{z}) = p(\mathbf{z}|\mathbf{x};\theta)$$

$$\sum_{\mathbf{z}} p(\mathbf{z}|\mathbf{x};\theta) \log \frac{p(\mathbf{x},\mathbf{z};\theta)}{p(\mathbf{z}|\mathbf{x};\theta)} = \sum_{\mathbf{z}} p(\mathbf{z}|\mathbf{x};\theta) \log \frac{p(\mathbf{z}|\mathbf{x};\theta)p(\mathbf{x};\theta)}{p(\mathbf{z}|\mathbf{x};\theta)}$$

$$= \sum_{\mathbf{z}} p(\mathbf{z}|\mathbf{x};\theta) \log p(\mathbf{x};\theta)$$

$$= \log p(\mathbf{x};\theta) \underbrace{\sum_{\mathbf{z}} p(\mathbf{z}|\mathbf{x};\theta)}_{=1}$$

$$= \log p(\mathbf{x};\theta)$$

- ELBO is a lower bound of log-likelihood $\log p_\theta(x;\theta)$
  - ① How tight is the bound?
  - ② How to achieve equality?

Variational Inference

- **Q1: How Tight is the Bound ELBO?**
  - Equality "log-likelihood = ELBO" holds when $q(z) = p(z \mid x; \theta)$

    ➔ Inequality due to $q(z) \neq p(z \mid x; \theta)$

  - Using KL Divergence to measure the gap:

$$D_{KL}(q(\mathbf{z}) \| p(\mathbf{z}|\mathbf{x}; \theta)) = -\sum_{\mathbf{z}} q(\mathbf{z}) \log p(\mathbf{z}, \mathbf{x}; \theta) + \log p(\mathbf{x}; \theta) - H(q) \geq 0$$

Re-arrange Terms

$$\log p(\mathbf{x}; \theta) = \mathrm{ELBO} + D_{KL}(q(\mathbf{z}) \| p(\mathbf{z}|\mathbf{x}; \theta))$$

Discrepancy Term

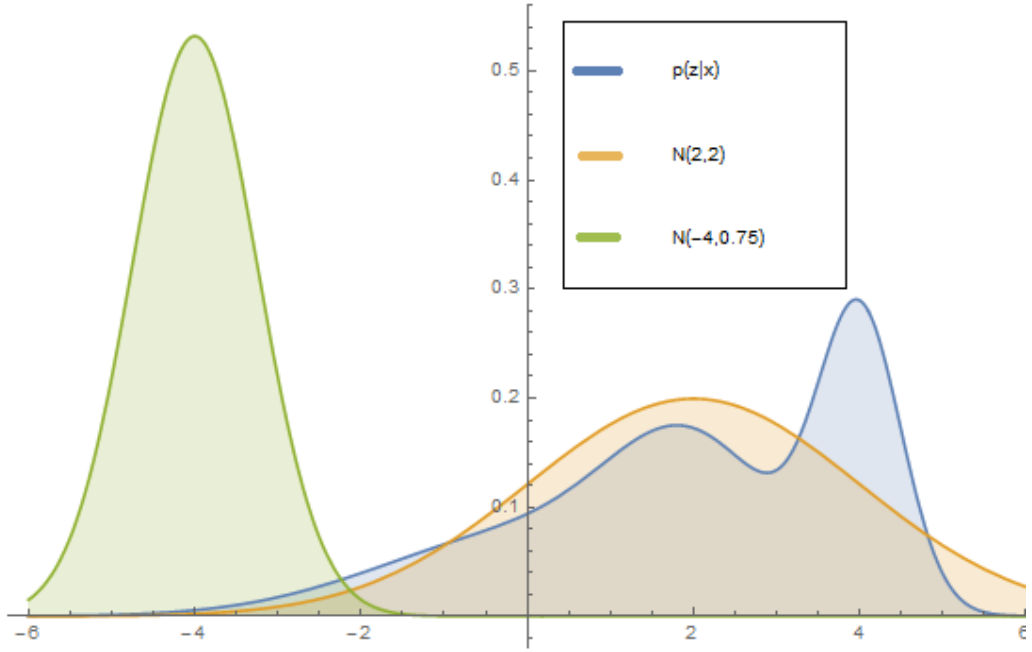
  - **The closer $q(z)$ is to $p(z \mid x; \theta)$, the closer ELBO is to the log-likelihood**

Variational Inference

- **Q2: How to achieve the equality $q(z) = p(z \mid x; \theta)$?**
  - **Motivation:** Approximate $p(z \mid x; \theta)$ with $q(z; \phi)$
    - ✓ $q(z; \phi)$ is a **tractable** distribution **parameterized** by ϕ
  - **Variational Inference:** Choose ϕ so that $q(z; \phi)$ is closest to $p(z \mid x; \theta)$
    - ✓ **变分推断**
    - ✓ **Inference (推断)**

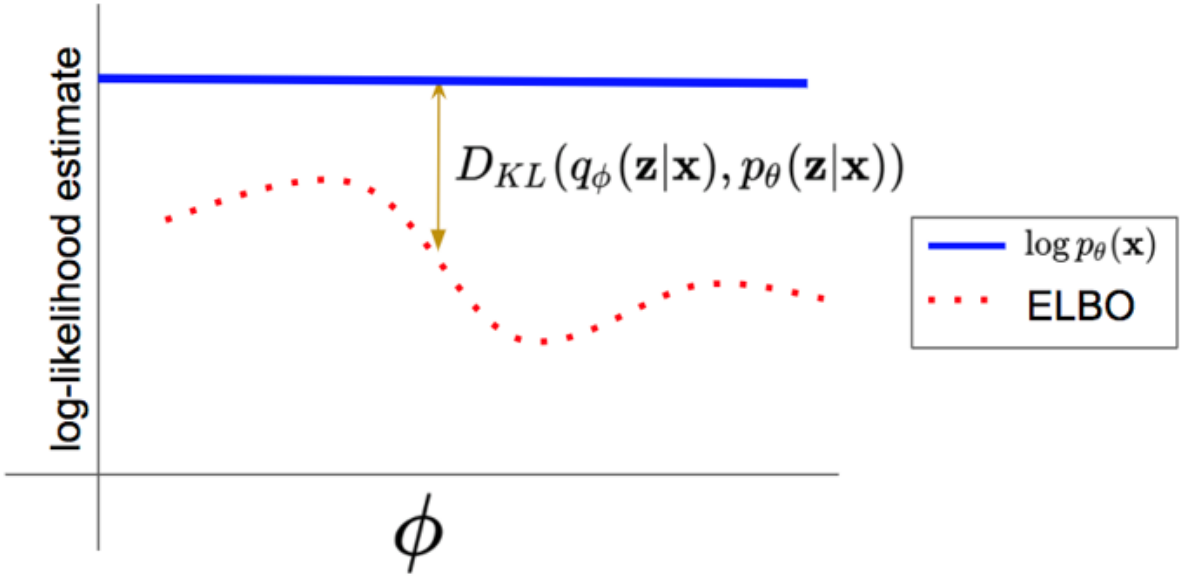
      Computing $p(z \mid x)$
    - ✓ **Variational (变分)**

      Approximate by optimizing ϕ



Variational Inference

- **Variational Parameterization of ELBO**



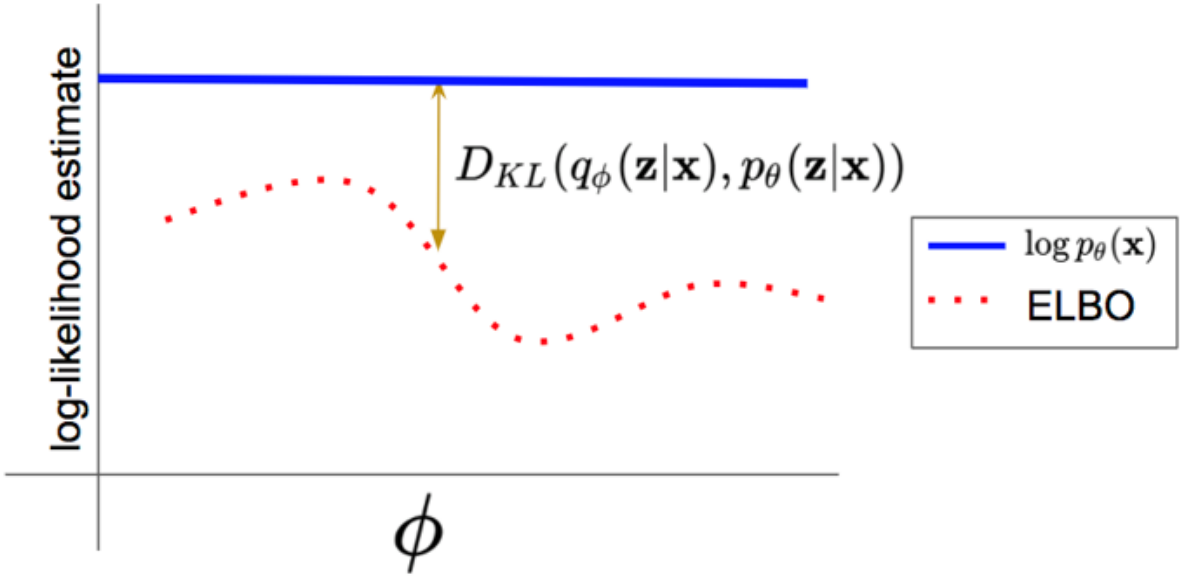
$$\log p(\mathbf{x};\theta) \geq \sum_{\mathbf{z}} q(\mathbf{z};\phi)\log p(\mathbf{z},\mathbf{x};\theta) + H(q(\mathbf{z};\phi)) = \underbrace{\mathcal{L}(\mathbf{x};\theta,\phi)}_{\text{ELBO}}$$

$$\log p(\mathbf{x};\theta) = \mathcal{L}(\mathbf{x};\theta,\phi) + D_{KL}(q(\mathbf{z};\phi)\|p(\mathbf{z}|\mathbf{x};\theta))$$

$q(z;\phi)$ approximate $p(z \mid x;\theta)$ better ⟺ $D_{KL}\big(q(z;\phi) \parallel p(z \mid x;\theta)\big)$ is smaller ⟺ ELBO is closer to $\log p(x;\theta)$

Variational Inference

- **Variational Parameterization of ELBO**



$$\log p(\mathbf{x};\theta) \geq \sum_{\mathbf{z}} q(\mathbf{z};\phi)\log p(\mathbf{z},\mathbf{x};\theta) + H(q(\mathbf{z};\phi)) = \underbrace{\mathcal{L}(\mathbf{x};\theta,\phi)}_{\text{ELBO}}$$

$$\log p(\mathbf{x};\theta) = \mathcal{L}(\mathbf{x};\theta,\phi) + D_{KL}(q(\mathbf{z};\phi)\|p(\mathbf{z}|\mathbf{x};\theta))$$

- **Question:** How to **jointly optimize over θ and ϕ** to maximize the ELBO over the training dataset $\mathcal{D}$?

Summary

- **Latent Variables Models**
  - Explicitly model the latent variable z via $p(x, z) = p(x \mid z)\, p(z)$
- **Pros**
  - Easy to build flexible models $p(x)$ using simple blocks $p(x \mid z)$
  - Suitable for unsupervised learning
- **Cons**
  - Hard to evaluate likelihoods
  - Hard to train via maximum-likelihood
  - Accurately estimating the posterior inference $p(z \mid x)$ is hard, thus requires variational approximation (i.e., variational inference)

Supplementary Material

- **Stanford CS 236 Lecture 5**
  - https://www.youtube.com/watch?v=MAGBUh77bNg&list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8&index=5
- **Cornell CS 6785 Lecture 5**
  - https://www.youtube.com/watch?v=KQOQyrkU8rk&list=PL2UML_KCiC0UPzjW9BjO-lW6dqliu9O4B&index=5
- **Deep Generative Models**
  - Chapter 5 Latent Variable Models
- **The Principle of Diffusion Models**
  - Chapter B.2 Variational Perspective: From VAEs to DDPMs